A negative answer to an eventual covering question for rational dilates

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Abstract

Let $E \subset (0, \infty)$ be measurable with positive Lebesgue measure. We ask whether, for almost every $x > 0$, every sufficiently large integer n admits an integer $r \geq 1$ such that

$$x \in \frac{r}{n} E.$$

The answer is negative. In fact, using a lemma of Buczolich and Mauldin, we construct a measurable set $E \subset (0, \infty)$ of positive measure and an interval $J \subset (0, \infty)$ such that for every $x \in J$ there are infinitely many integers n for which

$$x \notin \frac{r}{n} E \qquad \text{for every } r \in \mathbb{N}.$$

Thus the desired eventual covering property can fail on a whole interval.

1 Introduction

The problem under discussion is the following.

Question. *Let $E \subset (0, \infty)$ be a measurable set of positive Lebesgue measure. Is it true that, for almost all $x > 0$, for all sufficiently large integers n, there exists an integer $r \geq 1$ such that*

$$x \in \frac{r}{n} E?$$

If E contains a nondegenerate interval, then the answer is trivially positive. Indeed, if $(a, b) \subset E$ with $0 < a < b$, then for any fixed $x > 0$ the interval

$$\left(\frac{nx}{b}, \frac{nx}{a} \right)$$

has length $nx(\frac{1}{a} - \frac{1}{b}) > 1$ for all sufficiently large n, hence contains an integer $r \geq 1$; then $nx/r \in (a, b) \subset E$, equivalently $x \in \frac{r}{n} E$.

The purpose of this note is to show that positivity of measure alone does *not* suffice.

Theorem 1. *There exists a measurable set $E \subset (0, \infty)$ of positive Lebesgue measure and an interval*

$$J = \left[\frac{16}{25}, \frac{2}{3} \right]$$

such that for every $x \in J$ there are infinitely many integers $n \geq 1$ with the property that

$$x \notin \frac{r}{n} E \qquad \textit{for every } r \in \mathbb{N}.$$

In particular, the question above has a negative answer.

The proof is unconditional. Its only external input is a lemma of Buczolich and Mauldin, reproduced below in the exact form needed for our argument. The counterexample itself does not appear in their paper; it is obtained by repurposing their shell construction.

2 Notation and a lemma of Buczolich–Mauldin

For a set $A \subset (0, \infty)$ define

$$\Phi(A) := [1/2, 1) \cap \bigcup_{m \in \mathbb{N}} \frac{1}{m} A = \{x \in [1/2, 1) : \exists m \in \mathbb{N} \text{ such that } mx \in A\}.$$

Thus $x \in \Phi(A)$ precisely when some positive integer multiple of x belongs to A, subject to the restriction $x \in [1/2, 1)$.

We shall also use the intervals

$$I_F := (8/9, 1), \qquad I_\infty := \left[\frac{16}{25}, \frac{2}{3}\right].$$

The following lemma is the key input from Buczolich and Mauldin.

Lemma 2 (Buczolich–Mauldin)**.** *There exists $K_0 \in \mathbb{N}$ such that for every integer $k \geq K_0$ there exists an integer N_k with the following property: for every integer $\nu \geq N_k$ there is an open set*

$$H_{k,\nu} \subset (2^{\nu-1}, 2^\nu)$$

such that

$$I_\infty \subset \Phi(H_{k,\nu}),$$
$$\mathcal{L}(I_F \cap \Phi(H_{k,\nu})) < 5 \cdot 2^{-k}.$$

Remark 3. This is exactly the lemma used by Buczolich and Mauldin to prove their Theorem 1 on the convergence behavior of $\sum_{n=1}^{\infty} f(nx)$ for characteristic functions of open sets.

From Lemma 2 we immediately obtain a sequence of pairwise disjoint open sets with the same two properties.

Lemma 4. *There exists an integer $K \in \mathbb{N}$ and a sequence $(H_k)_{k \geq K}$ of pairwise disjoint open sets such that for every $k \geq K$:*

1. *$H_k \subset (2^{\nu_k - 1}, 2^{\nu_k})$ for some strictly increasing sequence of integers $(\nu_k)_{k \geq K}$;*
2. *$I_\infty \subset \Phi(H_k)$;*
3. *$\mathcal{L}(I_F \cap \Phi(H_k)) < 5 \cdot 2^{-k}$.*

Moreover, K can be chosen so large that

$$\sum_{k=K}^{\infty} 5 \cdot 2^{-k} < \mathcal{L}(I_F) = \frac{1}{9}. \tag{1}$$

Proof. Choose $K \geq \max\{K_0, 7\}$. For each $k \geq K$, select an integer $\nu_k \geq N_k$ in such a way that $\nu_K < \nu_{K+1} < \nu_{K+2} < \cdots$, and then choose an open set

$$H_k := H_{k,\nu_k} \subset (2^{\nu_k - 1}, 2^{\nu_k})$$

as given by Lemma 2. Since the dyadic shells $(2^{\nu_k-1}, 2^{\nu_k})$ are pairwise disjoint, the sets H_k are pairwise disjoint. Properties (2) and (3) are inherited directly from Lemma 2.

Finally,

$$\sum_{k=K}^{\infty} 5 \cdot 2^{-k} = 5 \cdot 2^{-K+1} \leq \frac{5}{64} < \frac{1}{9},$$

because $K \geq 7$. This proves (1). □

3 Construction of the counterexample

Fix K and $(H_k)_{k \geq K}$ as in Lemma 4, and set

$$F := \bigcup_{k=K}^{\infty} H_k.$$

Since F is a countable union of open sets, it is measurable. Define

$$E := I_F \setminus \Phi(F).$$

We now verify that this set E has the desired properties.

It will be convenient to introduce also the multiplicative saturation

$$M(E) := \bigcup_{r \in \mathbb{N}} rE.$$

Observe that for every $x > 0$, every $n \in \mathbb{N}$, and every $r \in \mathbb{N}$,

$$x \in \frac{r}{n} E \quad \Longleftrightarrow \quad nx \in rE. \tag{2}$$

Consequently,

$$\exists r \in \mathbb{N} : \; x \in \frac{r}{n} E \quad \Longleftrightarrow \quad nx \in M(E). \tag{3}$$

Lemma 5. *The set E has positive Lebesgue measure.*

Proof. By definition of $\Phi(F)$ and $F = \bigcup_{k \geq K} H_k$,

$$\Phi(F) = \bigcup_{k=K}^{\infty} \Phi(H_k).$$

Hence

$$I_F \cap \Phi(F) \subset \bigcup_{k=K}^{\infty} (I_F \cap \Phi(H_k)).$$

By countable subadditivity and Lemma 4,

$$\mathcal{L}(I_F \cap \Phi(F)) \leq \sum_{k=K}^{\infty} \mathcal{L}(I_F \cap \Phi(H_k)) < \sum_{k=K}^{\infty} 5 \cdot 2^{-k} < \frac{1}{9} = \mathcal{L}(I_F).$$

Therefore

$$\mathcal{L}(E) = \mathcal{L}\big(I_F \setminus \Phi(F)\big) = \mathcal{L}(I_F) - \mathcal{L}(I_F \cap \Phi(F)) > 0.$$

□

Lemma 6. *The sets F and $M(E)$ are disjoint:*

$$F \cap M(E) = \varnothing.$$

Equivalently, for every $r \in \mathbb{N}$,

$$F \cap rE = \varnothing.$$

Proof. Assume for contradiction that $y \in F \cap M(E)$. Then $y \in rE$ for some $r \in \mathbb{N}$, so we may write

$$y = rx \qquad \text{with } x \in E.$$

Since $E \subset I_F \subset [1/2, 1)$, we have $x \in [1/2, 1)$. On the other hand, $y \in F$ implies

$$x = \frac{y}{r} \in \frac{1}{r}F.$$

Therefore

$$x \in [1/2, 1) \cap \frac{1}{r}F \subset \Phi(F).$$

But $x \in E = I_F \setminus \Phi(F)$ by construction, a contradiction. Thus $F \cap M(E) = \varnothing$. □

Lemma 7. *For every $x \in I_\infty$ there are infinitely many integers $n \in \mathbb{N}$ such that*

$$nx \in F.$$

Consequently, for every $x \in I_\infty$ there are infinitely many integers $n \in \mathbb{N}$ such that

$$x \notin \frac{r}{n}E \qquad \textit{for every } r \in \mathbb{N}.$$

Proof. Fix $x \in I_\infty$. By Lemma 4(2), for each $k \geq K$ we have $x \in \Phi(H_k)$. By definition of $\Phi(H_k)$, there exists an integer $n_k \in \mathbb{N}$ such that

$$n_k x \in H_k \subset F.$$

Since the sets H_k are pairwise disjoint, the integers n_k are pairwise distinct: indeed, if $n_k = n_\ell$ for $k \neq \ell$, then the common value $n_k x$ would belong to both H_k and H_ℓ, impossible. Hence the set $\{n_k : k \geq K\}$ is infinite, proving the first assertion.

Now let $n = n_k$ be one of these integers. Then $nx \in F$. If there existed $r \in \mathbb{N}$ such that $x \in \frac{r}{n}E$, then by (2) we would have $nx \in rE \subset M(E)$, contradicting Lemma 6. Therefore no such r exists. Since there are infinitely many choices of $n = n_k$, the second assertion follows. □

Proof of Theorem 1. Let E be the measurable set constructed above. By Lemma 5, E has positive Lebesgue measure. By Lemma 7, every $x \in I_\infty = [16/25, 2/3]$ admits infinitely many integers n such that

$$x \notin \frac{r}{n}E \qquad \text{for all } r \in \mathbb{N}.$$

Therefore, on the whole interval I_∞, the eventual covering property from the question fails. In particular, it is not true that for every positive-measure set $E \subset (0, \infty)$ the property holds for almost every $x > 0$. □

4 Concluding remark

The counterexample is stronger than a mere failure on a set of positive measure: the exceptional set contains an entire interval. The role of the Buczolich–Mauldin construction is to produce disjoint "shells" H_k that every $x \in I_\infty$ hits infinitely often under integer multiplication, while the reciprocal shadows $\Phi(H_k)$ have summable measure inside I_F. Taking the complement of the total shadow inside I_F creates a positive-measure set E whose integer multiples avoid all of the shells. Each hit $nx \in H_k \subset F$ then becomes an obstruction to the desired representation $x \in (r/n)E$.

References

[1] Z. Buczolich and R. D. Mauldin, *On the convergence of* $\sum_{n=1}^{\infty} f(nx)$ *for measurable functions*, manuscript dated June 22, 2001.